

# Cover Letter

**Date:** July 27, 2026

**To:** The Editors *Genome Biology*

**Re:** Submission of “Delta Dependency Prioritizes Paralog-Based Synthetic Lethality Candidates Across Solid Tumor Types”

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Dear Editors,

We submit for consideration in *Genome Biology* our manuscript entitled “**Delta Dependency Prioritizes Paralog-Based Synthetic Lethality Candidates Across Solid Tumor Types.**” This work presents Delta Dependency (DD), a simple, univariate metric for paralog-based synthetic lethality (SL) prediction, supported by exploratory analyses across five orthogonal layers of evidence spanning genomic, proteomic, pharmacologic, clinical-stratification, and structural analyses.

## Key contributions:

1. **An interpretable metric matched to tiered, citation-verified gold standards.** On a curated set of 12 literature-derived paralog-SL pairs with tiered, citation-verified experimental evidence, DD achieves AUROC = 0.676 — robust to label quality (0.725 excluding two DepMap-derived pairs; unchanged under a direction-strict relabelling) — and both lineage-evaluable positives on the Tier A + Tier B external benchmark (direct or genotype-conditional experimental evidence) rank above all unlabeled controls (AUROC = 1.000; 2 of 5 benchmark pairs lineage-evaluable). In a head-to-head benchmark on the same 72 pairs (6 known positives), the interpretable composite score (AUROC = 0.831) matches the best of four multi-feature classifiers under leave-one-pair-out cross-validation (SVM-RBF 0.843, random forest 0.722; linear classifiers unstable at this sample size). Driver-mutation status follows gene-class-specific rules (TSG: likely loss-of-function; oncogene: hotspot), with the complete variant classification table provided as Supplementary Table S7 and a full directionality audit of the |DD| scoring convention released with the code.
2. **Pan-cancer scope: 23 solid tumor types, 7 CPTAC cohorts.** This is the largest systematic evaluation of paralog-SL across human cancers (66,595 HGNC paralog pairs  $\times$  1,208 DepMap cell lines). DD exceeds AUROC = 0.7 in 9 of 17 evaluable lineages. CPTAC proteomics across seven independent cohorts (BRCA, COAD, LUAD, GBM, PDAC, UCEC, LUSC;  $n = 771$  samples) confirms paralog co-variation at the protein level — EP300 $\leftrightarrow$ CREBBP significant in 5 of 7 cohorts — while the RNA-level signal is weak (AUROC = 0.547, near the random expectation of 0.5).
3. **PRISM drug sensitivity validates the framework.** A discovery-stage scan of 1,482 compounds identifies 633 compound–paralog associations with selective killing in driver-mutant cell lines (roughly 158 expected false positives at the permissive cutoff; the threshold is deliberately relaxed because this is a screen, not a confirmatory analysis). Known biology is recapitulated (MEK inhibitors  $\rightarrow$  KRAS-mutant, mTOR/AKT inhibitors  $\rightarrow$  PTEN-mutant, HDAC inhibitors  $\rightarrow$  EP300-mutant), and novel associations are uncovered.
4. **Clinical stratification and safety analysis.** Using the official DepMap MSIsensor2 annotation, MSI-H subgroups show paralog-SL signal at least as strong as MSS subgroups in both colorectal and endometrial lineages, mutation type modulates compensation strength (truncating  $>$  missense for the leading TSG pair), and a quantitative therapeutic window framework nominates ARID1A $\rightarrow$ ARID1B

(DWS = 2.82) as the leading selective candidate, with a biomarker-enriched trial design centered on truncating ARID1A mutations.

5. **Every number is one-command reproducible.** All analyses use publicly available DepMap and CPTAC data, and the method requires only subtraction: no GPU, no model training. Beyond the open-source R package (paraLogSL v1.1.0) and Python pipeline, every quantitative claim is verified by a layered audit system: per-script claim checks for headline metrics (16/16 verified), classifier benchmarks, and covariate-adjusted regressions (10/10 verified), plus a master closed-loop audit that independently recomputes **165 numeric claims** — every AUROC, p-value, sample size, and effect estimate across both the main text and the supplementary information — directly from the analysis artifacts and fails loudly on any mismatch. Every figure is generated by a script from the same artifacts (no manual edits; the few stochastic label-placement algorithms are seeded for reproducibility), and both documents passed a full-page visual review (page-by-page contact sheets included in the repository). A single entry point (`verify_all.sh`) runs all audits plus the test suite (31 tests) in under one minute. The submission-version code is archived on Zenodo (<https://doi.org/10.5281/zenodo.21502030>; R package: <https://doi.org/10.5281/zenodo.21502113>).

We believe this manuscript is well-suited for *Genome Biology* because it combines methodological simplicity with exceptionally broad validation, addresses a clinically important problem (SL-based cancer target discovery), and sets a high standard for computational reproducibility that the community can immediately build on. The pan-cancer CPTAC validation and PRISM drug selectivity analysis substantially extend the evidence base beyond prior computational SL studies.

All data are from public repositories (DepMap, cBioPortal, PDC, TCGA). All code is openly available: analysis pipeline at <https://github.com/tjogzt/paralog-sl-predictor> and R package at <https://github.com/tjogzt/paralogSL>.

**Competing interests:** The authors declare no competing interests.

**Author contributions:** Q.Q.M. and T.Z. conceived the study. T.Z. developed the methodology, performed all computational analyses, and developed the R package. Q.Q.M. curated clinical datasets and provided gynecological oncology domain expertise. Q.Q.M. and T.Z. wrote the manuscript.

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We confirm that this manuscript is original, has not been published previously, and is not under consideration by any other journal. Both authors have read and approved the final manuscript and agree with its submission to *Genome Biology*.

We thank you for your consideration and look forward to your response.

Sincerely,

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**Suggested Reviewers:**

1. Dr. Francisca Vazquez — DepMap / Broad Institute (CRISPR dependency expertise)
2. Dr. Michael P. Snyder — Stanford (proteogenomics, CPTAC expertise)
3. Dr. Jason Moffat — University of Toronto (synthetic lethality screening)
4. Dr. Rameen Beroukhi — Dana-Farber / Broad Institute (cancer genomics, paralog dependencies)
5. Dr. Bing Zhang — Baylor College of Medicine (CPTAC proteogenomics)

**Opposed Reviewers:** None.